

Unit 3, Lesson 5: Rational Numbers in Real-World Context



Benchmark

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite directions using rational numbers. Compare them on a number line and explain the meaning of zero within its context.



Learning Targets

- ☐ I can determine positive and negative values in relation to real-world contexts.
- ☐ I can plot negative rational values given a real-world context.



3.5.1 Warm-Up: Mathematician Katherine Johnson

Born in 1918, Katherine Johnson was always fascinated by numbers. Graduating from high school at 14 and from college



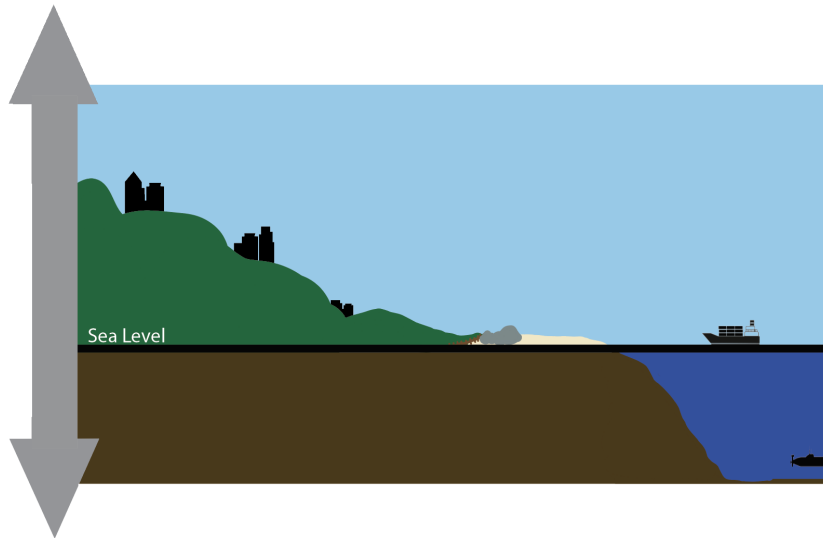
at 18, she went on to become one of the first African American mathematicians with the job title of “computer” at NASA, where she worked for three decades. As a computer, she personally checked the calculations made by electronic computers. Katherine calculated the launch window for America’s first human space flight. Her calculations were critical to the success of the Apollo Moon landing program and the start of NASA’s Space Shuttle program. In 2015, she received the Presidential Medal of Freedom from President Obama. In its official announcement, the White House cited Katherine’s influence on every major space program from Mercury to the Shuttle Program.

1. What stands out to you the most about Katherine Johnson?



3.5.2 Collaboration: Exploring Elevation

The picture shows different locations relative to sea level.



1. Describe how the picture can be used to explain the word *elevation*.

2. Ask two classmates for their description of the word elevation. Write their description in the table. *You are the only person who should write in the first column of the table.* Have your classmate initial next to your summary of their description stating that your summary is correct.

My Summary of _____'s Description of Elevation	Initials
My Summary of _____'s Description of Elevation	Initials

3. Using the picture, name one thing that is below sea level.
4. Name one thing that is above sea level.

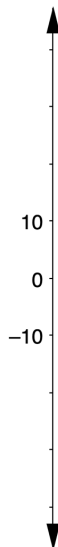


3.5.3 Guided Instruction: Positive and Negative Values in the Real World

Elevation is the vertical distance a location or object is from a given level, such as sea level.

Elevation is represented using positive and negative values.

1. If sea level is equivalent to an elevation of 0 units, then what type of value is an elevation that is below sea level?
2. Write each situation as an integer. Plot it on the vertical number line.
 - a. A hiker is 37 feet above sea level.
 - b. A diver is 19 feet below sea level.
 - c. How are these two integers related?
 - d. What does 0 represent in this situation?



3. Samuel L. Gravely, Jr. pioneered the way with a plethora of firsts for African Americans in the military. Gravely was the first African American Navy Vice Admiral, first African American to command a Navy warship, first African American to command a warship during combat, first African American to command a Navy fleet, and first African American to obtain Flag Rank in the military. If the Navy fleet included a submarine that was submerged 780 feet below sea level, which students' interpretation of the situation is correct? Circle the name of the student with the correct answer. For any statement that has an error, write a corrected statement.

Tianna	Keagan
The depth of the submarine is -780 feet below sea level.	780 feet below sea level can be represented by the integer -780 .

A bank **deposit** is the amount of money a person gives the bank for safekeeping in their account.

A bank **withdrawal** is the amount of money removed from a bank account.

An **expense** is a cost or the amount of money spent.

4. The following people have money in their bank accounts. Determine the positive and negative values that represent the effect each situation has on their account balance.

Real-World Context	Value
Paul paid \$190.90 in taxes.	
Marie made a deposit of \$50.	
Tyler made a withdrawal of \$200.	
Hannah gets paid an income of \$500 per week that she deposits in her account.	
Kaden pays \$758.91 in monthly expenses.	



3.5.4 Collaboration: Above or Below Zero

1. Represent each change in temperature on the number line. Answer the question for each situation.

At noon, the temperature was 5 degrees Celsius. By late afternoon, the temperature had risen by 6 degrees Celsius.		What was the temperature late in the afternoon?
The temperature was 8 degrees Celsius at midnight. By dawn, the temperature had dropped by 12 degrees Celsius.		What was the temperature at dawn?
Water freezes at 0 degrees Celsius, but the freezing temperature can be lowered by adding salt to the water. A student discovered that adding half a cup of salt to a gallon of water lowers its freezing temperature by 7 degrees Celsius.		What is the freezing temperature of the gallon of salt water?

Numbers below 0 are useful in contexts other than temperature.

2. With your partner, discuss and list other examples where numbers below zero can be used to represent a situation.



3.5.5 Try It: Representing the Real World With Positive and Negative Values

1. In a bank account, every time money is added or taken out of the account, a new balance is calculated. What does each of the following mean?
 - a. an account balance of $-\$23.56$
 - b. an account balance of $\$143.21$
 - c. an account balance of $\$0$

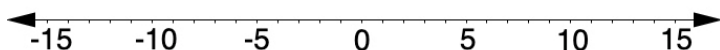
2. The change in position of the ball during each play of a football game is measured in yards.

a. What integer best represents a gain of 5 yards?

b. What integer best represents a loss of 15 yards?

c. What does the number 0 represent in this context?

d. Plot all three values on the number line.

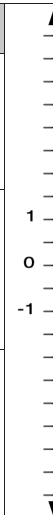


3. The average monthly rainfall for a city was calculated based on records kept over the past 50 years. The chart shows the difference between the average monthly rainfall and the rainfall amount this year for these months.

Month	Difference From Monthly Average (in inches)
January	$-2\frac{3}{4}$
February	$1\frac{5}{8}$
March	0
April	$-\frac{3}{8}$
May	$4\frac{1}{4}$

Explain what the value in the chart means for the following months in the context of this problem. Then, plot and label each month's value on the number line.

Month	Interpretation
February	
March	
April	





3.5.6 Your Turn!

1. Write a value that represents each description.

- a. an expense of \$40.23 _____
- b. a deposit of \$225.89 _____
- c. 16,000 feet above sea level _____
- d. a temperature increase of 14°F _____
- e. a withdrawal of \$271.49 _____
- f. a gain of 17 yards _____

2. Read each statement and select the correct description of each situation.

a. A diver is 60 feet below the surface of the ocean.

- ☐ The depth of the diver is 60 feet below the ocean's surface.
- ☐ The diver is -60 feet below the ocean's surface.

b. The elevation of the bottom of a boat with respect to sea level is given as -5 feet.

- ☐ The bottom of the boat is 5 feet above sea level.
- ☐ The bottom of the boat is 5 feet below sea level.



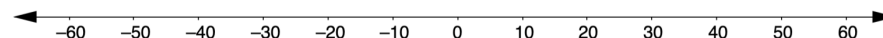
3.5.7 Wrap-Up: Ticket Out the Door

Keisha observes that one location is 55.2 feet above sea level and another is 46.8 feet below sea level.

1. What rational number represents a location of 55.2 feet above sea level?

2. What rational number represents a location of 46.8 feet below sea level?

3. Plot and label each rational number on the number line.



4. What does zero represent in this situation?